[redacted]

The “Simple Linear Regression” model can be written in the form:

$Y=b_0+b_1X$

1.1. In your own words, define each part of this simplified equation:

1.1.1. b_0

Answer here:

1.1.2. b_1

Answer here:

1.2. Now, imagine x represents **daily exercise** (minutes/day) and y represents **life expectancy** (years). Would you expect a positive or negative slope in this model? Briefly explain why.

Answer here:

1.3. Suppose a researcher fit a simple linear regression and found the equation:

$y = 60 + 0.25x$

where x is daily exercise (minutes/day) and y is life expectancy (years).

1.3.1. What does the intercept 60 represent in everyday terms?

Answer here:

1.3.2. What does the slope of 0.25 indicate about how life expectancy changes as daily exercise increases?

Answer here:

1.3.3. Using this model, predict the life expectancy of someone who exercises **for 40 minutes** daily.

Answer here:

[redacted]

2.1. In the “Fitting Linear Regression” infographic, you may notice that the regression model now includes an error/residual term ϵ:

$y = 0+1x +$

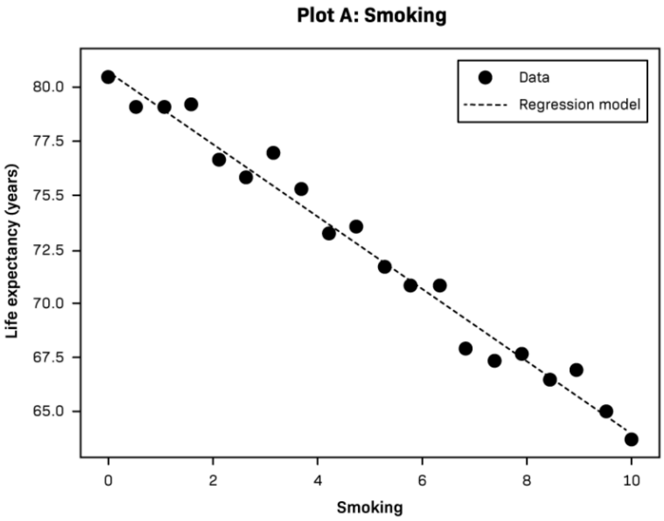
What does $\boldsymbol{\epsilon}$ represent in the context of “how well our model fits the data?”

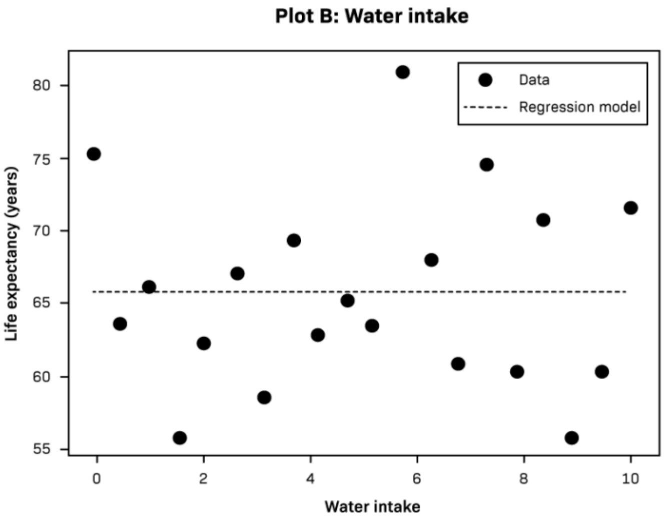
Answer here:

2.2. The infographic also introduces the concept of **R2**, the coefficient of determination. In your own words, what does R2 tell us about a linear regression model's performance?

Answer here:

[redacted]





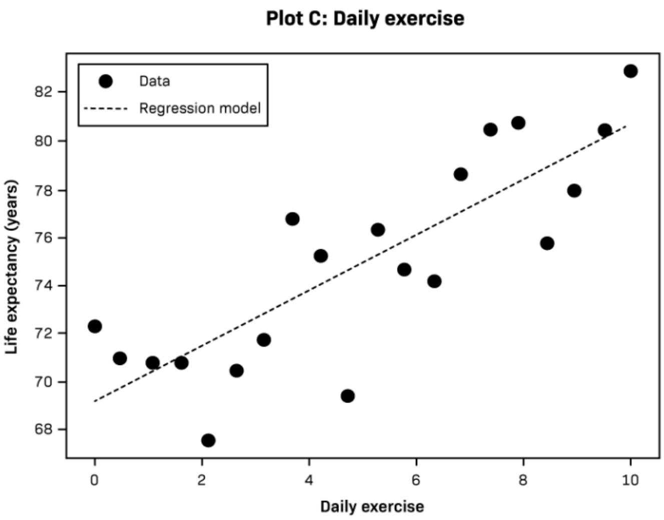


Image description: *Three scatter plots comparing life expectancy (y-axis, ranging from approximately 55 to 82 years) with different health factors (x-axis, scaled 0 to*

10). Each plot includes data points and a dashed regression line illustrating a potential linear relationship. On plot A, the health factor on the x-axis is smoking. Data points start high on the y-axis at low x-values and trend downward as x-values increase, with a regression line that slopes downward from left to right. The data points closely follow the regression line. On plot B, the health factor is water intake. Data points are scattered across the graph with no clear pattern, and a regression line remains nearly horizontal across all x-values. On plot C, the health factor is daily exercise. Data points begin lower on the y-axis at low x-values and trend upward as x-values increase, with a regression line that slopes upward from left to right. The data points are scattered around the regression line but follow the upward trend.

3.1. **R2 ranking:** Rank the 3 plots from highest to lowest R2. What clues support your ranking?

Answer here:

3.2. **Biological reasoning:** One of the plots shows a particularly strong relationship. In a real-world health context, what might explain such a marked trend?

Answer here:

[redacted]

You’ve learned how **R2**can indicate how closely a regression line fits the data. Here, you have a **simple** model (already fitted) relating **smoking rate** to **life expectancy** for several regions. After computing **R2** for this single-factor approach, you’ll consider how adding **daily exercise** as a **second** predictor could change the model’s explanatory power.

Below is a scatter plot relating **smoking rate** (x-axis, in % of adults) to **life expectancy** (y-axis, in years). The dashed line shows the fitted model:

y = 85 -0.5(smoking rate)

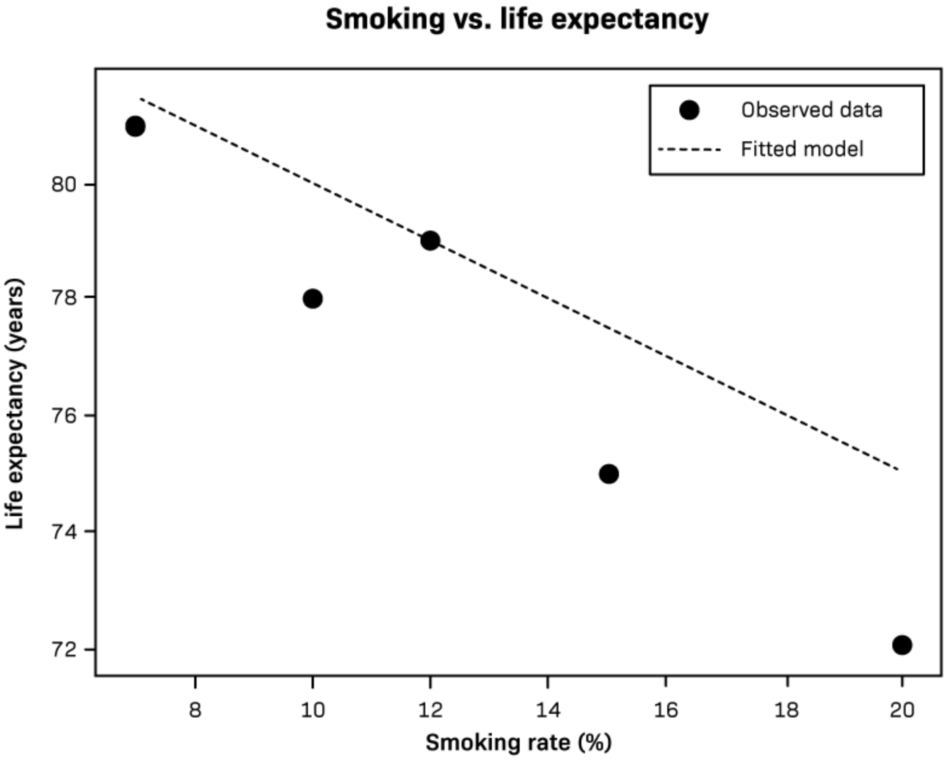


Image description: *A scatter plot titled "Smoking vs. life expectancy." The x-axis is labeled "Smoking rate" and runs from 6% to 20%. The y-axis is labeled "Life expectancy (years)" and runs from 72 to 82. A legend identifies observed data (black dots) and fitted model (dashed line). The plot contains 5 black data points that generally trend downward as smoking rate increases. A gray dashed line represents the fitted linear regression model, sloping downward from left to right.*

4.1. From the scatter plot, extract the necessary data to compute the R2 of this model.

Answer here:

4.2. In plain language, what does your R^2 value suggest about how smoking rate alone explains variation in life expectancy for these points?

Answer here:

4.3. Now, imagine you also have data on daily exercise (minutes/day) for each region. By adding that second predictor,

y = 85 -0.5(Smoking Rate) +exercise(Daily Exercise) ,

you might see a higher R^2.

Why might adding daily exercise help to increase R^2?

Answer here:

4.4. Provide one reason why adding more variables to a regression model might not always improve its usefulness or accuracy in real-world settings.

Answer here: